

# ASHISH KOSANA

Software Engineer, Backend and Full-Stack

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## SUMMARY

Backend and full-stack software engineer. I build and ship features end to end, from Amazon Web Services (AWS) infrastructure as code and REST APIs secured with Cognito to cross-platform Flutter and TypeScript user interfaces, deployed to production with typed tests that pass in Continuous Integration (CI). Strong in data structures, algorithms, and systems fundamentals, and I use LLMs to build and ship faster.

## SKILLS

- **Languages:** Python, TypeScript, JavaScript, Dart, SQL, Bash, Linux
- **Backend and APIs:** REST, HTTP, JSON API design, FastAPI, SQLAlchemy, pydantic v2, OAuth 2.0, JSON Web Tokens (JWT), Stripe, hexagonal (ports and adapters) architecture, caching, microservices
- **Cloud and Infrastructure (AWS):** Lambda, API Gateway, Cognito, CDK for Infrastructure as Code (IaC), EventBridge, Secrets Manager, IAM, S3
- **Databases:** DynamoDB (NoSQL), PostgreSQL, SQLite (relational database) **Containers:** Docker, docker-compose
- **Frontend:** Next.js, React (server-side rendered Progressive Web App, PWA), Flutter (cross-platform iOS, Android)
- **Testing:** pytest, unit and integration testing, test doubles and fakes, mypy --strict, ruff, evaluation harness (precision/recall/F1)
- **Workflow:** Git, pull request (PR) and code review, Continuous Integration/Continuous Deployment (CI/CD) with GitHub Actions, Agile
- **Core CS:** data structures, algorithms, Object-Oriented Programming (OOP), operating systems, computer networks

## EXPERIENCE

**Software Engineer (Intern, full-time conversion track)** | Crewtron | Tempe, AZ | May 2026 - Present

- Shipped clock-in/out with geofence auto-tracking and a live crew map end to end, using a Flutter UI, Python on AWS Lambda, and DynamoDB. Crew visibility is scoped by role, and presence triggers auto clock-in.
- Designed REST APIs secured with AWS Cognito and provisioned the backend as code with AWS CDK and scoped IAM policies. Built merchant review-request flows over email/SMS.
- Built cross-platform Flutter features on the Stripe API (invoicing, quotes, payments) and a UTC timezone pipeline that drives a timezone-aware calendar.
- Improved quality with structured QA, OWASP Top 10 security reviews, and a consistent Git/PR workflow. Authored an API integration-test suite reaching **~82% coverage** and integrated LLM-based automation into internal tooling.

## PROJECTS

**Rythu | Telugu-first farmer weather and crop-advisory PWA (deployed, real users)** | Next.js, TypeScript, Python, AWS Lambda, DynamoDB, API Gateway, CDK, Amplify  
[github.com/Ashishkosana/rythu](https://github.com/Ashishkosana/rythu)

- Ships a server-rendered Telugu PWA (Next.js on AWS Amplify, git-push CI/CD) over a Python Lambda and API Gateway backend, with infrastructure as code in AWS CDK. A rule-based advisory engine uses GPS and OpenStreetMap geocoding and applies government-verified fertilizer rules that account for nutrient overlap and prevent over-application.
- Caches third-party weather forecasts in DynamoDB with a TTL, so repeat lookups skip the external weather API, cutting latency and cost. **~135 tests** (pytest and Vitest), mypy --strict, CI green.

**ledgerline | exactly-once payments service (pure backend/systems, no LLM)** | Python, FastAPI, SQLAlchemy, PostgreSQL, Docker  
[github.com/Ashishkosana/ledgerline](https://github.com/Ashishkosana/ledgerline)

- Prevents double-charges on client retries with storage-layer idempotency (a unique key committed in the same transaction as the charge) over an explicit payment state machine, and records every money movement in an append-only double-entry ledger with an automated reconciliation job.
- Solves the dual-write problem with a transactional outbox, a relay, and consumer-side dedup (effectively-once delivery), plus a dead-letter queue and exponential backoff. Benchmark: **zero double-charges under 200 concurrent identical requests, ~1,460 payments/sec.** 27 tests, mypy --strict.

**tick | durable job & cron scheduler on Postgres (pure backend/systems, no LLM)** | Python, SQLAlchemy, PostgreSQL  
[github.com/Ashishkosana/tick](https://github.com/Ashishkosana/tick)

- Concurrent-safe job claiming with SELECT ... FOR UPDATE SKIP LOCKED, leases with crash recovery (a dead worker's job is reclaimed), retries with backoff, a dead-letter state, and recurring schedules. Benchmark: **500 jobs across 10 workers processed exactly once, zero collisions.** 14 tests, mypy --strict.

## EDUCATION

**Bachelor of Science (B.S.) in Computer Science** | University of Massachusetts Lowell | Graduated Dec 2025

GPA 3.51/4.00 | Dean's List | Transferred from SR Institute of Technology (JNTU), Hyderabad to complete the B.S.